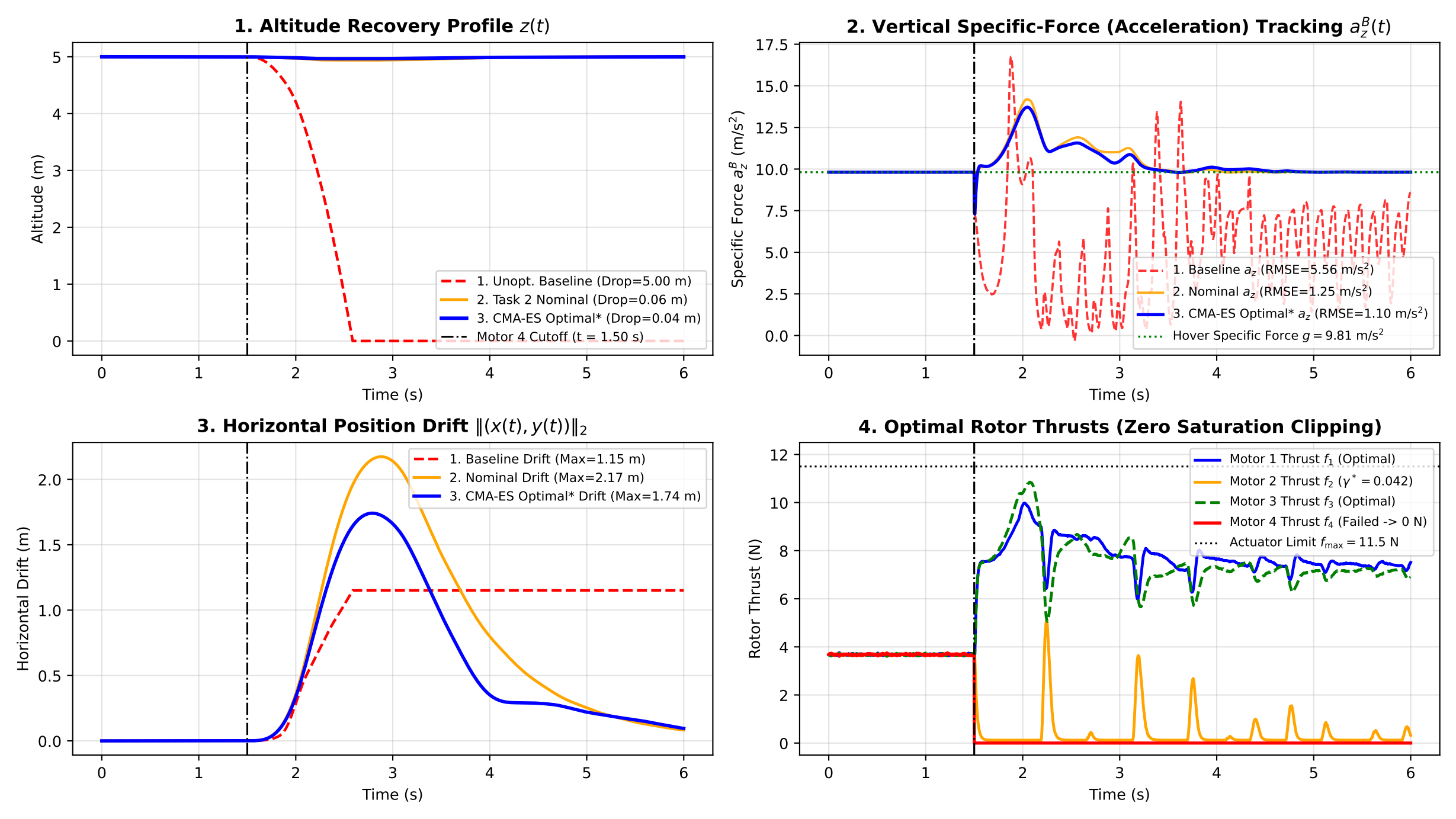

1. Altitude Recovery Profile $z(t)$
Altitude (m)
Time (s)
1. Unopt. Baseline (Drop=5.00 m)
2. Task 2 Nominal (Drop=0.06 m)
3. CMA-ES Optimal* (Drop=0.04 m)
Motor 4 Cutoff (t = 1.50 s)
2. Vertical Specific-Force (Acceleration) Tracking $a_z^B(t)$
Specific Force a_z^B (m/s^2)
Time (s)
1. Baseline a_z (RMSE=5.56 m/s^2)
2. Nominal a_z (RMSE=1.25 m/s^2)
3. CMA-ES Optimal* a_z (RMSE=1.10 m/s^2)
Hover Specific Force $g = 9.81$ m/s^2
3. Horizontal Position Drift $\|(x(t), y(t))\|_2$
Horizontal Drift (m)
Time (s)
1. Baseline Drift (Max=1.15 m)
2. Nominal Drift (Max=2.17 m)
3. CMA-ES Optimal* Drift (Max=1.74 m)
4. Optimal Rotor Thrusts (Zero Saturation Clipping)
Rotor Thrust (N)
Time (s)
Motor 1 Thrust f_1 (Optimal)
Motor 2 Thrust f_2 ($\gamma^* = 0.042$)
Motor 3 Thrust f_3 (Optimal)
Motor 4 Thrust f_4 (Failed -> 0 N)
Actuator Limit $f_{max} = 11.5$ N